

Unleashing the Power of Python in Autodesk Maya

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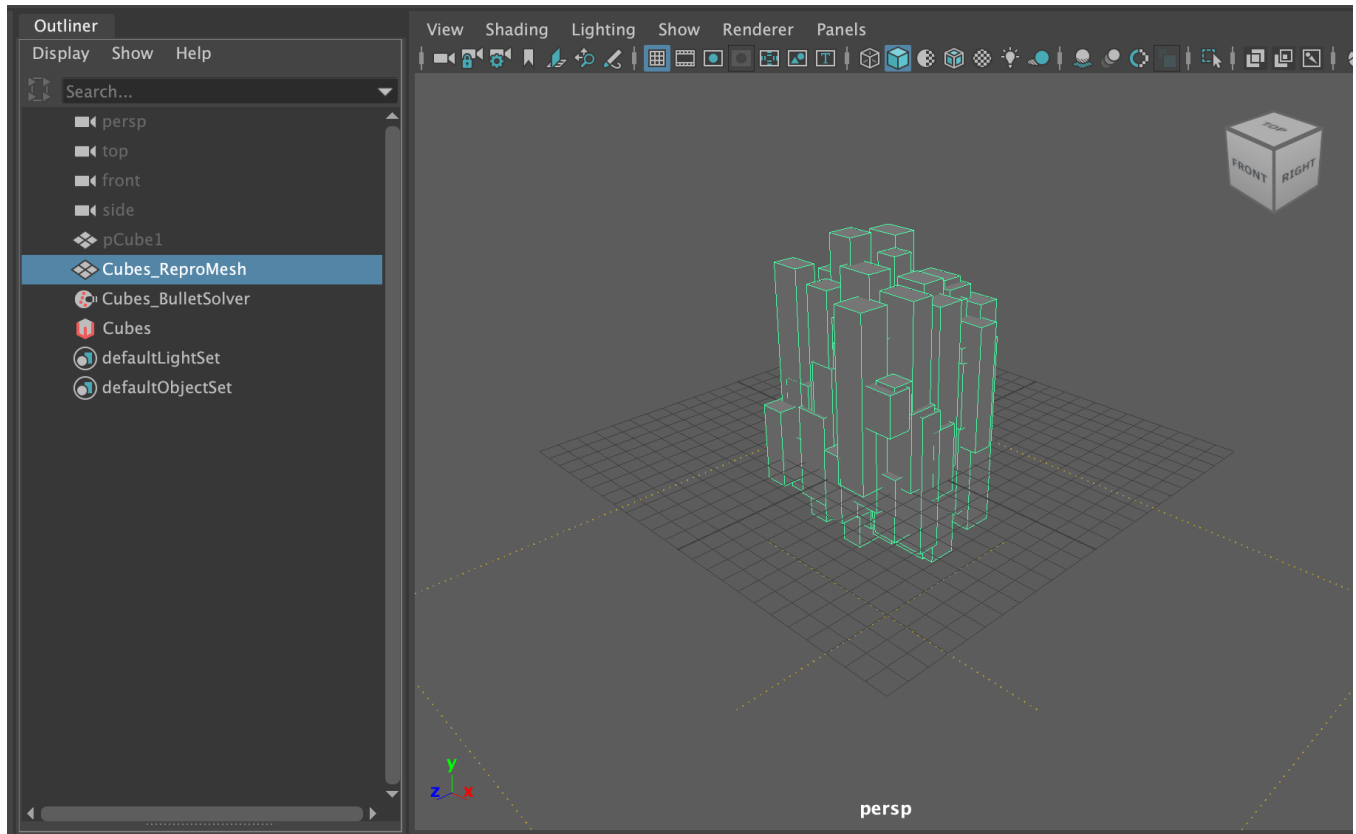


Figure 1: MASH networks combined with python scripting provides a powerful framework for automatically generating geometry.

ABSTRACT

Coding empowers automation. Scripts can handle mundane and repetitive tasks in an efficient and precise manner. This course will offer will use an hands-on interactive format to walk attendees through representative scripting projects, selected to be useful for everyday workflows. It is intended to be an intermediate course. The goal is to cover provide enough information for attendees to build on later. Python scripting can automate many tasks in Maya, from running simple commands to developing plug-ins. Attendees

will learn how to automate a simple task using the magic of scripting, through a hands-on projects. The course will placing objects in a scene through scripting MASH (motion graphic) networks. Attendees should walk away with a solid understanding of the power Python scripting and Maya commands provide, and the the ability to conceive their own advance projects for Maya. Attendees should have programming experience, preferably in Python, but a solid grasp of the foundational programming constructs should suffice. Attendees should have Autodesk Maya, Python, and Visual Studio Code pre-loaded on their devices if they intend to follow along.

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1 INTRODUCTION

MASH can be used to create “versatile motion design animations with procedural node networks”. Effects can be quickly assembled in a custom manner, and can be chained to realize unique variations.

2 PROGRAMMING MASH NETWORKS

This course will develop a script that creates a MASH network to generate some rectangular shapes of random length and adds effects to them so when the user presses play dynamic simulation is invoked. Figures ?? and ?? illustrate the progression of the dynamic animation. The real benefit of such a network is experienced through viewing the actual animation. Time-permitting, attendees will have the opportunity to experiment with different types of geometry, and animation settings.

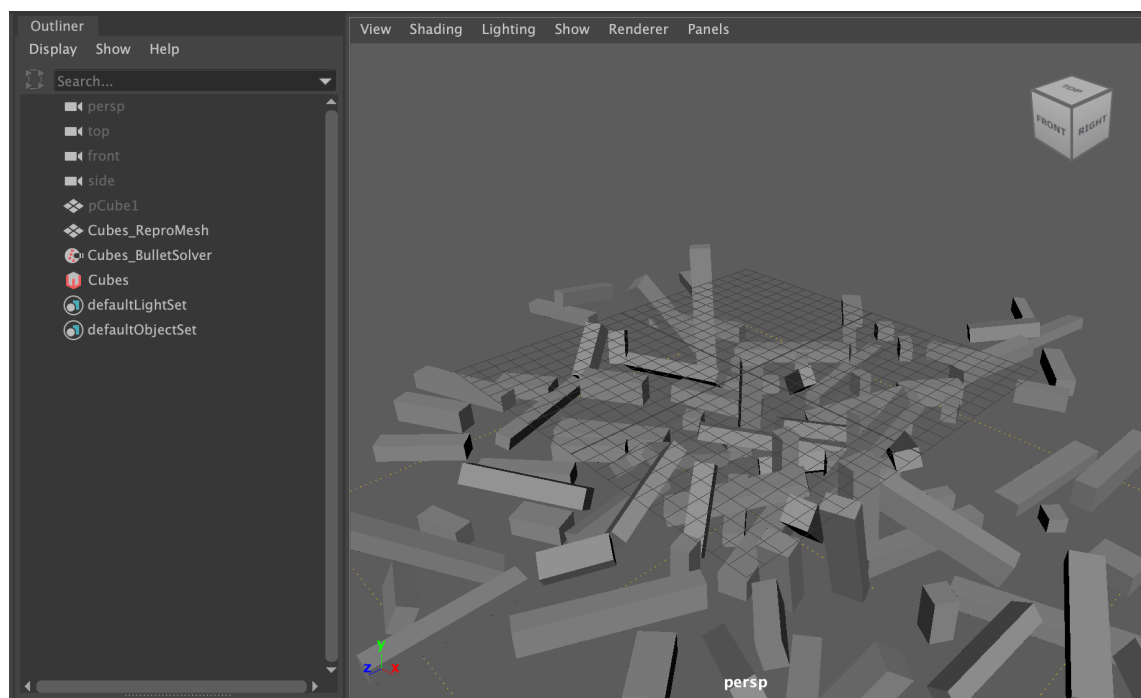


Figure 2: The dynamic simulation applied to random collection of rectangular shapes generated using a MASH network

3 SUMMARY

MASH networks are a powerful tool to quickly generate repeating elements, dynamic processes and stochastic simulations. Attendees will learn how to embed Python scripts to inject more autonomy into your MASH networks.